

Predicting Employee Flight Risk

Moving from Reactive Attrition to Proactive Retention using Machine Learning

Fernando Pellerano



<https://www.linkedin.com/in/fernandopellerano/>



The True Cost of Turnover is a Systemic Financial Drain

15% - 20%

Of an employee's salary is spent strictly on recruiting a replacement.

\$7,645

The average direct financial cost to hire just one new employee.

**52 Days &
2.5% Revenue**

Average time to fill a position, alongside the 1-2.5% of total revenue lost while bringing a new hire up to speed.

Managers spend up to 40% of working hours on tasks that do not generate direct revenue, such as hiring.
High turnover is a compound fracture to the bottom line.

The Predictive Hypothesis: Data-Driven Retention



Applying machine learning classification algorithms to historical HR data will allow us to accurately predict which employees are at high risk of resigning, enabling proactive retention to avoid significant financial losses.



Diagnostic Inputs (Features)

- ✓ Job Satisfaction
- ✓ Work-Life Balance
- ✓ Education Level
- ✓ Performance Rating
- ✓ Job Involvement



Target Output (Binary Classification)

0 (Stable / Stays with the company)

1 (Flight Risk / Resigns)

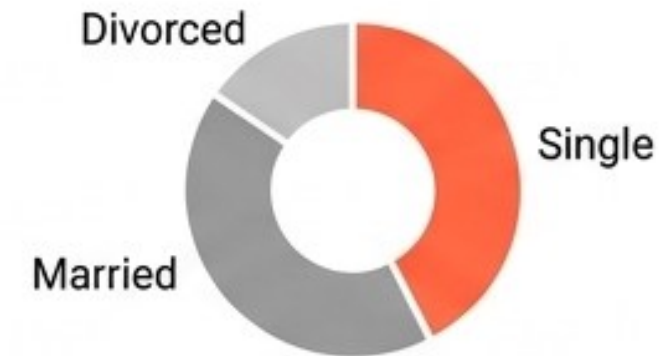
Profiling the Flight Risk: Symptoms of Resignation

Job Role Dynamics



Sales Representatives leave at a significantly higher rate than other roles.

Marital Status



Single individuals exhibit a distinctly higher flight risk.

Career Stage



Low Job Level / Inexperienced employees are heavily represented in the attrition cohort.

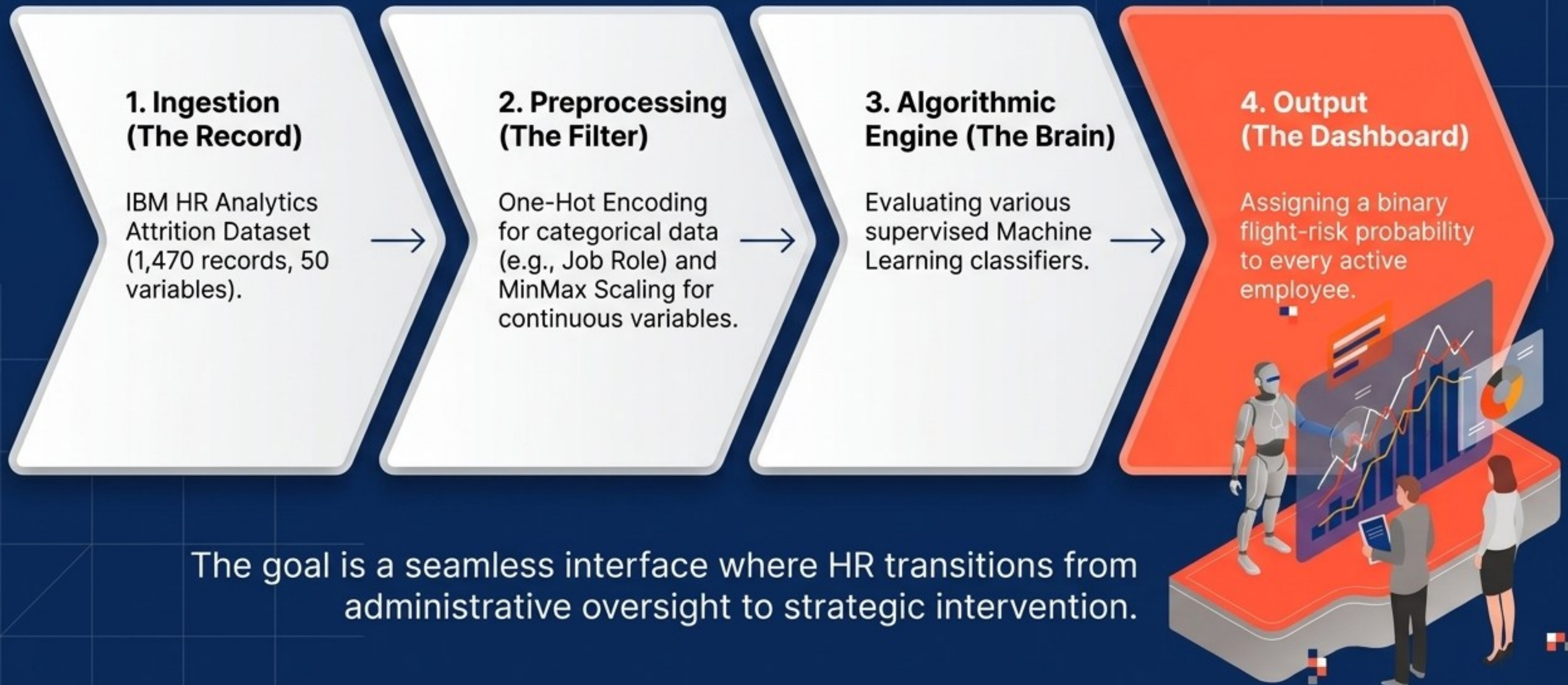
Engagement Deficit



Low Job Involvement correlates heavily with sudden resignation.

Synthesis: Attrition is not random; it clusters around highly specific demographic and role-based profiles.

Solution Architecture: From Raw Data to Proactive Action



Defining Success: The Accuracy Paradox

In our dataset, the vast majority stay (Class 0), and a minority leave (Class 1).

If a model simply guesses "everyone stays", it achieves ~84% accuracy, but is 100% useless to the business.

		PREDICTED CLASS	
		STAY (0)	LEAVE (1)
ACTUAL CLASS	STAY (0)	TRUE NEGATIVES (TN) (Correct Stay)	FALSE POSITIVES (FP) (False Alarm)
	LEAVE (1)	FALSE NEGATIVES (FN) (Missed Resignation)	TRUE POSITIVES (TP) (Caught Flight Risk)

TRANSLATION MATRIX

Metric:
Accuracy

Overall correct predictions.

Misleading. A false sense of security.

Metric:
Precision

True Positives / Total Predicted Positives.

When we flag a flight risk, how often are we right? (Minimizes false alarms)

Metric: **Recall**
(The Target Metric)

True Positives / Actual Positives.

Of all the people who will actually leave, how many did we catch? (Maximizes retention opportunities)

Metric: **F1-Score**

Harmonic mean of Precision and Recall.

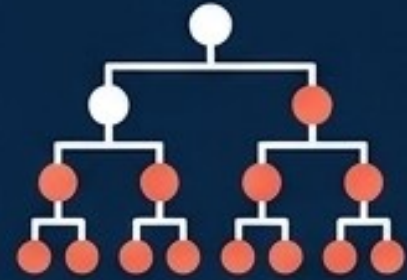
The ultimate balancing act between false alarms and missed resignations.

The Arsenal: Evaluating Predictive Algorithms



1. Logistic Regression

A linear approach estimating the **probability of a binary outcome**. The simplest, most interpretable baseline.



2. Random Forest Classifier

An **ensemble of decision trees** trained on **random subsets** of data. Excellent for handling complex, non-linear relationships.



3. CatBoost

An advanced **gradient boosting** algorithm specifically optimized to handle **categorical data** (like Job Roles and Departments) seamlessly.



4. Deep Learning (ANN)

A multi-layered artificial **neural network** (3 hidden layers of 500 neurons) designed to uncover highly complex, hidden patterns in the data.

The Model Showdown: Why Complexity Fails the Business

Model	Test Accuracy	Recall (Yes Class)	F1-Score	Overfitting Risk
Deep Learning	85%	34%	0.43	Severe (100% Train vs 85% Test)
CatBoost	86.1%	25%	0.37	High (Gap ~0.11)
Random Forest	85.0%	29%	0.38	High
Baseline Logistic Regression	89.6%	40%	0.50	Low



Insight: The complex models fall into the Accuracy Paradox. They optimize for the majority class (people who stay) and completely fail to detect the actual flight risks (Recall under 35%).

The Verdict: Adjusted Logistic Regression Wins

By implementing ``class_weight='balanced'``, we force the algorithm to penalize missed resignations heavily, prioritizing Recall without relying on over-complex architectures.

Before and After

Before (Baseline LR)

- Accuracy: 89.67%
- Flight Risks Caught: 19 out of 48
- Recall: 40%

**+ 11 Additional
Employees Caught**

After (Adjusted LR)

- Accuracy: **76.09%**
- Flight Risks Caught: **30 out of 48**
- Recall: **62%**

The Business Translation: By trading a 13% drop in generic accuracy, the business gains a massive 55% improvement in identifying real flight risks. We caught 11 additional at-risk employees who would have previously slipped through the cracks.

The Prognosis: From Diagnostic to ROI

Every employee successfully retained through early intervention saves the company a minimum of \$7,645 and protects 52 days of lost productivity.

1. Strategic Deployment

The Adjusted Logistic Regression model provides the best trade-off between early detection (62% Recall) and avoiding false alarms, without the overfitting risks of deep learning.

2. The New Standard

HR is no longer a reactive, administrative function. Equipped with this precision dashboard, HR leadership can surgically deploy retention budgets to the right people before the resignation letter is even written.

Thank you

Fernando Pellerano



<https://www.linkedin.com/in/fernandopellerano/>